

Basic Concepts of Probability

Union

- $A \cup B = \{x : x \in A \vee x \in B\}$
- $\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n = \{x : x \in A_1 \vee x \in A_2 \vee \dots \vee x \in A_n\}$

Intersection

- $A \cap B = \{x : x \in A \wedge x \in B\}$
- $\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n = \{x : x \in A_1 \wedge x \in A_2 \wedge \dots \wedge x \in A_n\}$

Complement $A' = \{x : x \in S \wedge x \notin A\}$

Mutually Exclusive / Disjoint $A \cap B = \emptyset$

Event Operations

- $A \cap A' = \emptyset$
- $A \cap \emptyset = \emptyset$
- $A \cup A' = S$
- $(A')' = A$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup B = A \cup (B \cap A')$
- $A = (A \cap B) \cup (A \cap B')$

De Morgan's Law

- $(A_1 \cup A_2 \cup \dots \cup A_n)' = A_1' \cap A_2' \cap \dots \cap A_n'$
- $(A_1 \cap A_2 \cap \dots \cap A_n)' = A_1' \cup A_2' \cup \dots \cup A_n'$

Permutation Selection and arrangement of r objects out of n : $P_r^n = \frac{n!}{(n-r)!}$

Combination Selection of r objects out of n without regards to order:

$$C_r^n = \binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad \binom{n}{r} = \binom{n}{n-r}$$

Probability Axioms

- $0 \leq P(A) \leq 1$
- $P(S) = 1$
- $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$

Basic Properties of Probability

- $P(\emptyset) = 0$
- For mutually exclusive events,
 $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$
- $P(A') = 1 - P(A)$
- $P(A) = P(A \cap B) + P(A \cap B')$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- If $A \subset B$ then $P(A) \leq P(B)$

Conditional Probability The probability of B given A occurring is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Multiplication Rule $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$ given $P(A) \neq 0$ or $P(B) \neq 0$

Inverse Probability Formula $P(A|B) = \frac{P(A)P(B|A)}{P(B)}$

Independence $P(A \cap B) = P(A)P(B)$, denoted $A \perp B$

Some Properties of Independence

- If $P(A) > 0, P(B) > 0, A \perp B$, then A and B are not mutually exclusive
- If $P(A) > 0, P(B) > 0$, and they are mutually exclusive, then $A \not\perp B$
- S and $\emptyset$ are independent of any other event
- If $A \perp B$, then $A \perp B', A' \perp B, A' \perp B'$

Partition For mutually exclusive events with $\bigcup_{i=1}^n A_i = S, A_1, A_2, \dots, A_n$ is a partition of S

Law of Total Probability For a partition of S ,

$$P(B) = \sum_{i=1}^n P(B \cap A_i) = \sum_{i=1}^n P(A_i)P(B|A_i)$$

Baye's Theorem For a partition of $S, P(A_k|B) = \frac{P(A_k)P(B|A_k)}{\sum_{i=1}^n P(A_i)P(B|A_i)}$

Random Variables

Probability Mass Function for Discrete Random Variables

$$f(x) = \begin{cases} P(X = x) & x \in R_X \\ 0 & x \notin R_X \end{cases}$$

- $f(x_i) \geq 0$ for all $x_i \in R_X$
- $f(x) = 0$ for all $x \notin R_X$
- $\sum_{i=1}^{\infty} f(x_i) = 1$

Probability Density Function for Continuous Random Variables

- $f(x) \geq 0$ for all $x \in R_X, f(x) = 0$ for all $x \notin R_X$
- $\int_{R_X} f(x)dx = 1$
- For $a \leq b, P(a \leq X \leq b) = \int_a^b f(x)dx$

It suffices to check conditions 1 and 2

Cumulative Distribution Function $F(x) = P(X \leq x)$

- Discrete RV: $F(x) = \sum_{t \in R_X, t \leq x} P(x = t),$
 $P(a \leq X \leq b) = F(b) - F(a^-)$
- Continuous RV: $F(x) = \int_{-\infty}^x f(t)dt, f(x) = \frac{dF(x)}{dx},$
 $P(a \leq X \leq b) = F(b) - F(a)$

Expectation / Mean

- Discrete RV: $\mu_X = E(X) = \sum_{x_i \in R_X} x_i f(x_i)$
- Continuous RV: $\mu_X = E(X) = \int_{-\infty}^{\infty} x f(x)dx = \int_{x \in R_X} x f(x)dx$

Properties of Expectation

- $E(aX + b) = aE(X) + b$
- $E(X + Y) = E(X) + E(Y)$
- For arbitrary $g(\cdot)$
 - Discrete RV: $E[g(X)] = \sum_{x \in R_X} g(x)f(x)$
 - Continuous RV: $E[g(X)] = \int_{R_X} g(x)f(x)dx$

Variance and Properties $\sigma_X^2 = V(X) = E(X - \mu_X)^2$

- Discrete RV: $V(X) = \sum_{x \in R_X} (x - \mu_X)^2 f(x)$
- Continuous RV: $V(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)dx$
- $V(X) \geq 0$ for any X , with equality holding iff $P(X = E(X)) = 1$
- $V(aX + b) = a^2 V(X)$
- Alternative Formula: $V(X) = E(X^2) - [E(X)]^2$

Standard Deviation $\sigma_X = \sqrt{V(X)}, \sigma_X \geq 0$

Joint Distributions

Discrete Joint Probability Function $f_{X,Y}(x, y) = P(X = x, Y = y),$

$(x, y) \in R_{X,Y}$

- $f_{X,Y}(x, y) \geq 0, (x, y) \in R_{X,Y}$
- $f_{X,Y}(x, y) = 0, (x, y) \notin R_{X,Y}$
- $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} f_{X,Y}(x_i, y_j) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} P(X = x_i, Y = y_j) = 1$ or
 $\sum \sum_{(x,y) \in R_{X,Y}} f(x, y) = 1$
- $P((X, Y) \in A) = \sum \sum_{(x,y) \in A} f(x, y)$

Continuous Joint Probability Function

$P((X, Y) \in D) = \int \int_{(x,y) \in D} f_{X,Y}(x, y)dydx,$ or

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y)dydx$$

- $f_{X,Y}(x, y) \geq 0, (x, y) \in R_{X,Y}$
- $f_{X,Y}(x, y) = 0, (x, y) \notin R_{X,Y}$
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)dxdy = 1$ or $\int \int_{(x,y) \in R_{X,Y}} f_{X,Y}(x, y)dxdy = 1$

Marginal Probability Distribution Projection of 2D function onto 1D function

- Y Discrete: $f_X(x) = \sum_y f_{X,Y}(x, y)$
- Y Continuous: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dy$

Conditional Distribution Distribution of Y given that X is observed,

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

- Defined for $f_X(x) > 0$

2. If $f_X(x) > 0$, then $f_{X,Y}(x, y) = f_X(x)f_{Y|X}(y|x)$

3. $P(Y \leq y|X = x) = \int_{-\infty}^y f_{Y|X}(y|x)dy$

4. $E(Y|X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)dy$

Independent Random Variables $X_1, X_2, \dots, X_n$ are independent iff

$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2)\dots f_{X_n}(x_n).$ $R_{X,Y}$ must be a product space of $R_X \times R_Y$ by satisfying all possibilities.

Properties of Independent Random Variables

- $P(X \in A; Y \in B) = P(X \in A)P(Y \in B)$, for any real $x, y,$
 $P(X \leq x; Y \leq y) = P(X \leq x)P(Y \leq y)$
- For arbitrary $g_1(\cdot)$ and $g_2(\cdot), g_1(X)$ and $g_2(Y)$ are independent
- If $f_X(x) > 0$, then $f_{Y|X}(y|x) = f_Y(y)$

Checking Independence

- $R_{X,Y}$ where the probability function is positive must be a product space
- For $(x, y) \in R_{X,Y}, f_{X,Y}(x, y) = C \times g_1(x) \times g_2(y)$, i.e. it can be factorised into terms independent of other variables

Marginal Distribution under Independence Given $g_1(x)$ and $g_2(y),$

- X Discrete: Probability mass function is $f_X(x) = \frac{g_1(x)}{\sum_{t \in R_X} g_1(t)}$
- X Continuous: Probability density function is $f_X(x) = \frac{g_1(x)}{\int_{t \in R_X} g_1(t)dt}$

Expectation

- Discrete: $E(g(X, Y)) = \sum_x \sum_y g(x, y)f_{X,Y}(x, y)$
- Continuous: $E(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f_{X,Y}(x, y)dydx$

Covariance $cov(X, Y) = E[(X - E(X))(Y - E(Y))]$ using

$g(X, Y) = (X - E(X))(Y - E(Y))$

- Discrete: $cov(X, Y) = \sum_x \sum_y (x - \mu_X)(y - \mu_Y)f_{X,Y}(x, y)$
- Continuous: $cov(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y)f_{X,Y}(x, y)dxdy$

Properties of Covariance

- $cov(X, Y) = E(XY) - E(X)E(Y)$
- If X and Y are independent, $cov(X, Y) = 0$, and $E(XY) = E(X)E(Y)$
- $cov(aX + b, cY + d) = ac \cdot cov(X, Y)$ using
 $cov(X + b, Y) = cov(X, Y)$ and $cov(aX, Y) = acov(X, Y)$
- $V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2ab \cdot cov(X, Y)$ using
 $V(X + Y) = V(X) + V(Y) + 2cov(X, Y)$

Properties of Variance and Covariance

- For independent X and $Y, V(X \pm Y) = V(X) + V(Y)$
- $V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n) + 2 \sum_{j>i} cov(X_i, X_j)$
- For independent $X_1, X_2, \dots, X_n,$
 $V(X_1 \pm X_2 \pm \dots \pm X_n) = V(X_1) + V(X_2) + \dots + V(X_n)$

Special Probability Distributions

Discrete Distributions

Discrete Uniform Distribution The random variable assumes discrete values with equal probability

- pmf: $1/k$ if there are k values it can take on

$$\mu_X = E(X) = \frac{1}{k} \sum_{i=1}^k x_i$$

$$\sigma_X^2 = V(X) = E(X^2) - [E(X)]^2 = \frac{1}{k} \sum_{i=1}^k x_i^2 - \mu_X^2$$

Bernoulli Trial Random experiment with only two possible outcomes

Bernoulli Random Variable $X \sim \text{Bernoulli}(p)$ Success of a Bernoulli trial

- pmf: $p, x = 1; 1 - p = q, x = 0$
- $f_X(x) = p^x(1 - p)^{1-x}, x = 0, 1$
- $\mu_X = E(X) = p$
- $\sigma_X^2 = V(X) = p(1 - p) = pq$

Bernoulli Process Sequence of repeatedly performed independent and identical Bernoulli trials

Binomial Random Variable $X \sim \text{Bin}(n, p)$ Counts number of successes in n trials of a Bernoulli Process

- $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, 0 \leq x \leq n$
- $E(X) = np$
- $V(X) = np(1-p)$

Negative Binomial Distribution $X \sim NB(k, p)$ Let X be the number of independent and identically distributed Bernoulli trials needed until the k th success occurs. Then X follows a Negative Binomial Distribution.

- pmf: $f_X(x) = P(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, x \geq k$
- Derived from $k-1$ success in first $x-1$ trials and success on x th trial
- $E(X) = \frac{k}{p}$
- $V(X) = \frac{(1-p)k}{p^2}$

Geometric Distribution $X \sim \text{Geom}(p)$ Let X be the number of independent and identically distributed Bernoulli trials needed until the first success occurs. Then X follows a geometric distribution.

- pmf: $f_X(x) = P(X = x) = (1-p)^{x-1} p$
- $E(X) = \frac{1}{p}$
- $V(X) = \frac{1-p}{p^2}$

Poisson Random Variable $X \sim \text{Poisson}(\lambda)$ X denotes the number of events occurring in a fixed period of time or fixed region, λ is the expectation.

- pmf: $f_X(k) = P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, k \geq 0, k$ is the number of occurrences
- $E(X) = \lambda$
- $V(X) = \lambda$
- If $X \sim \text{Poisson}(\lambda_1), Y \sim \text{Poisson}(\lambda_2), X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2), (X|X+Y=n) \sim \text{Bin}(n, p = \frac{\lambda_1}{\lambda_1 + \lambda_2})$

Poisson Process A continuous time process where we count number of occurrences within some interval of time. The defining properties of a Poisson process with rate parameter α are

- the expected number of occurrences in an interval of length T is αT
- there are no simultaneous occurrences
- the number of occurrences in disjoint time intervals are independent

The number of occurrences in any interval T of a Poisson process follows a $\text{Poisson}(\alpha T)$ distribution

Poisson Approximation to Binomial Let $X \sim \text{Bin}(n, p)$. Suppose $n \rightarrow \infty$ and $p \rightarrow 0$ such that $\lambda = np$ remains constant. Then $X \sim \text{Poisson}(np)$. The approximation is good when $n \geq 20, p \leq 0.05$, or $n \geq 100, np \leq 10$

Continuous Distributions

Continuous Uniform Distribution $X \sim U(a, b)$

- pdf: $\frac{1}{b-a}, a \leq x \leq b$
- $E(X) = \frac{a+b}{2}$
- $V(X) = \frac{(b-a)^2}{12}$
- cdf: $\frac{x-a}{b-a}, a \leq x \leq b$

Exponential Distribution $X \sim \text{Exp}(\lambda)$

- pdf: $f_X(x) = \lambda e^{-\lambda x}, x \geq 0, \lambda > 0$
- $E(X) = \frac{1}{\lambda}$
- $V(X) = \frac{1}{\lambda^2}$
- cdf: $F_X(x) = 1 - e^{-\lambda x}, x \geq 0$
- $P(X > x) = e^{-\lambda x}, x > 0$
- Alternatively, $\mu = 1/\lambda, f_X(x) = \frac{1}{\mu} e^{-x/\mu}, E(X) = \mu, V(X) = \mu^2, F_X(x) = 1 - e^{-x/\mu}$
- $P(X > s + t | X > s) = P(X > t)$, i.e. the distribution is "memoryless"

Normal Distribution $X \sim N(\mu, \sigma^2)$

- pdf: $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$

- $E(X) = \mu$
- $V(X) = \sigma^2$
- The total area under the curve is 1

- Two curves with the same σ^2 have identical shapes, but different centers if they have different μ
- As σ increases, the curve flattens, and vice versa
- $X + Y \sim (\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

Standardised Normal Distribution Given $X \sim N(\mu, \sigma^2)$, let $Z = \frac{X-\mu}{\sigma}$.

- $Z \sim N(0, 1)$, pdf: $f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{z^2}{2})$
- For $P(x_1 < X < x_2)$, transform it to z_1, z_2 . Then $P(x_1 < X < x_2) = P(z_1 < Z < z_2)$
- $\phi(\cdot)$ and $\Phi(\cdot)$ denote pdf and cdf of the standard normal
- Then $P(x_1 < X < x_2) = \Phi(z_2) - \Phi(z_1)$
- $P(Z \geq 0) = P(Z \leq 0) = \Phi(0) = 0.5$
- $\Phi(z) = P(Z \leq z) = P(Z \geq -z) = 1 - \Phi(-z)$
- $Z \sim N(0, 1) \Rightarrow -Z \sim N(0, 1)$
- $Z \sim N(0, 1) \Rightarrow \sigma Z + \mu \sim N(\mu, \sigma^2)$

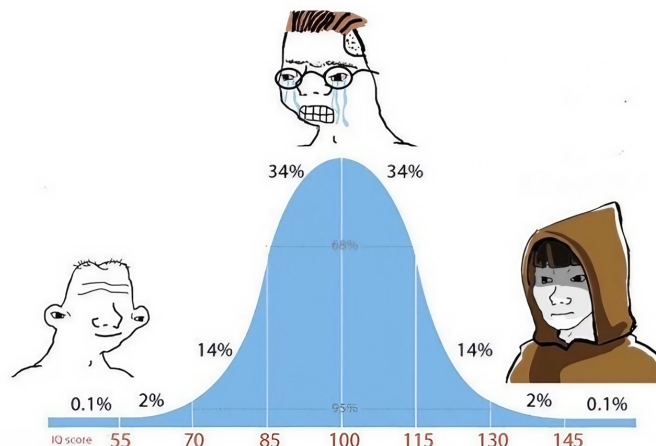
Quantile The α th (upper) quantile, $0 \leq \alpha \leq 1$, of X , is the number x_α such that $P(X \geq x_\alpha) = \alpha$. For Z , the α th (upper) quantile or 100α percentage point is $P(Z \geq z_\alpha) = \alpha$.

Normal Approximation to Binomial Let $X \sim \text{Bin}(n, p), E(X) = np, V(X) = npq$. As $n \rightarrow \infty, Z = \frac{X-E(X)}{\sqrt{V(X)}} = \frac{X-np}{\sqrt{np(1-p)}}$ is approximately $\sim N(0, 1)$, when $np > 5$ and $nq > 5$

Continuity Correction Accounts for discreteness when approximating to normal, basically when equal, shift outwards, when not equal, shift inwards

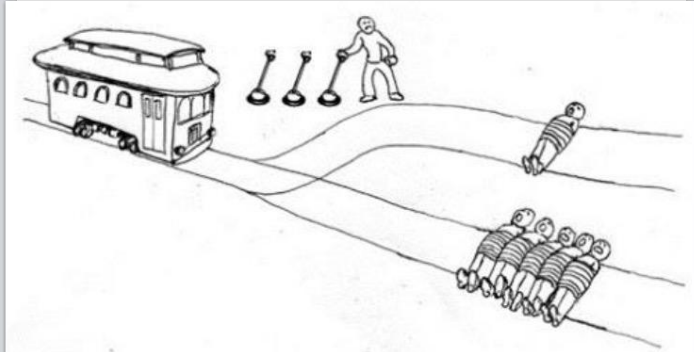
- $P(X = k) \approx P(k - 0.5 < X < k + 0.5)$
- $P(a \leq X \leq b) \approx P(a - 0.5 < X < b + 0.5)$
- $P(a < X < b) \approx P(a + 0.5 < X < b - 0.5)$

Bell Curve



Monty Hall Problem

Trolling with trolleys: the trolley problem goes full monty.



The Trolley Hall Problem

There are three levers. Two of the levers have no effect. You may only choose one lever to pull. You made your choice but before you pulled your lever, one of the other two levers is revealed as fake. Do you switch your choice?

Skewed Curve

